```
from scipy.stats import norm,uniform,poisson,binom
import matplotlib.pyplot as plt
import numpy as np
```

T1

$$E(Z) = E(X+Y) = E(X) + E(Y) = 5 + 5 = 10$$
$$Var(Z) = Var(X+Y) = Var(X) + Var(Y) = 4 + 4 = 8$$

T2

```
5-(48**0.5)/2
```

```
1.5358983848622456
```

```
# T2
for size in [10,100,1000,10000]:
    z = norm.rvs(loc=5,scale=2,size=size) + uniform.rvs(loc=5-(48**0.5)/2,scale=48**0.5,size=size)
    print(f'{size} times')
    print(f'mean: {z.mean()}')
    print(f'var: {z.var()}')
```

```
10 times
mean: 10.165881908468652
var: 10.075065038764489
100 times
mean: 9.943293155010156
var: 7.061017800675631
1000 times
mean: 10.11828774597905
var: 7.3472219813973805
10000 times
mean: 9.959345154028584
var: 7.8473231196869495
```

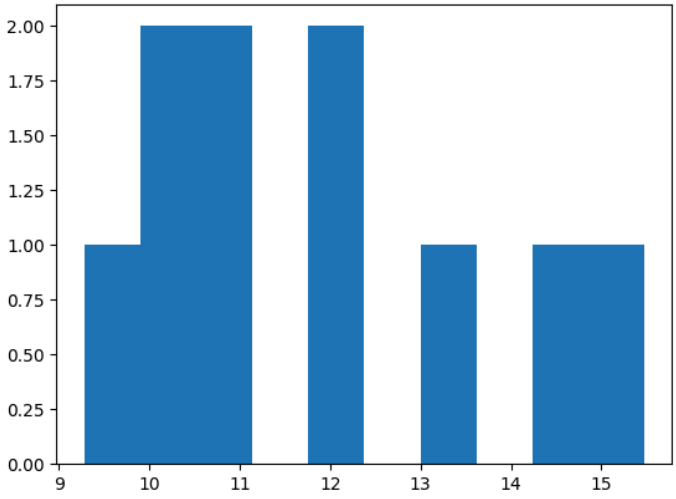
T3

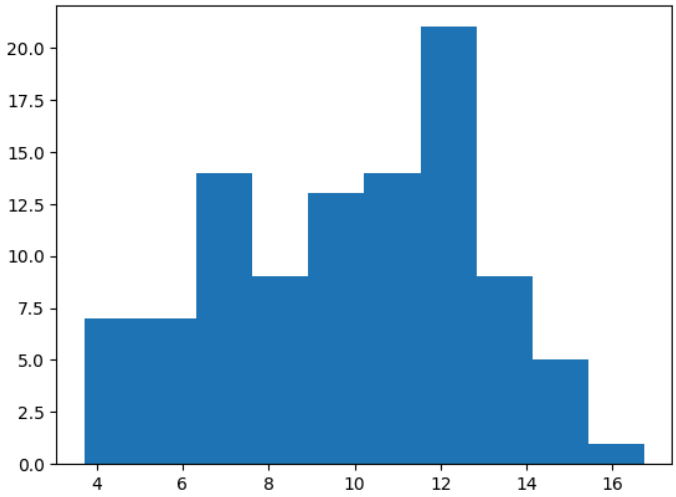
```
# T3
for size in [10,100,1000,10000]:
    count = 0
    for i in range(1000):
        z = norm.rvs(loc=5,scale=2,size=size) + uniform.rvs(loc=5-(48**0.5)/2,scale=48**0.5,size=size)
        mean = z.mean()
        if np.abs((mean-10)/10)<=0.01:count+=1
    print(f'{size} times: {count}')
```

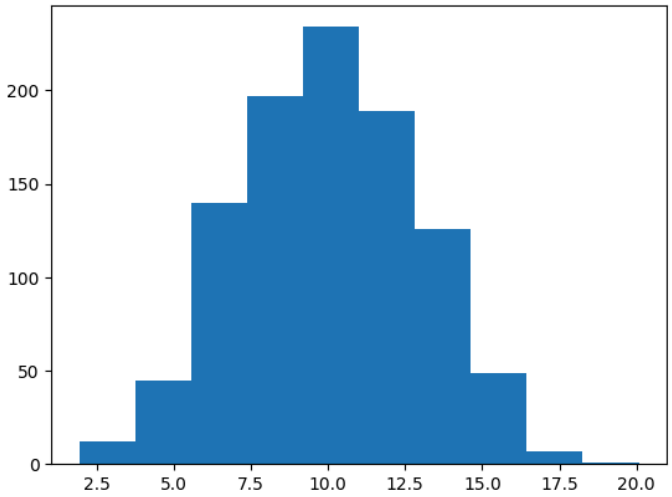
```
10 times: 93
100 times: 271
1000 times: 723
10000 times: 998
```

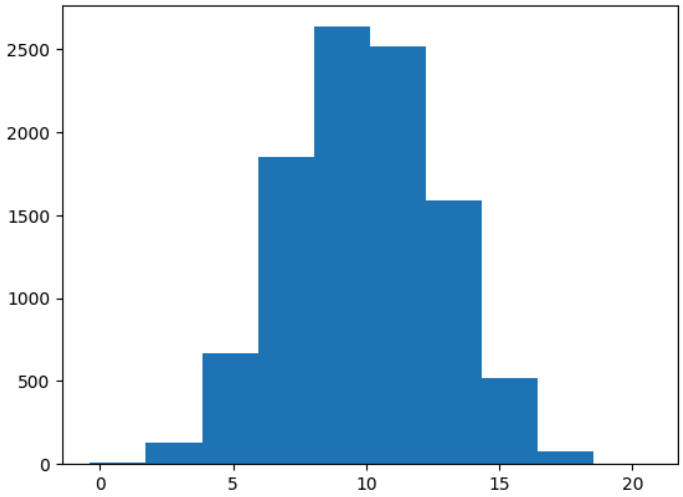
T4

```
# T4
for size in [10,100,1000,10000]:
    z = norm.rvs(loc=5,scale=2,size=size) + uniform.rvs(loc=5-(48**0.5)/2,scale=48**0.5,size=size)
    plt.hist(z)
    plt.show()
```


2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
9
10
11
12
13
14
15


20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
4
6
8
10
12
14
16


200
150
100
50
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


2500
2000
1500
1000
500
0
0
5
10
15
20

T5

What does the Law of Large Numbers say about your winnings if you make a large number of 1-dollar bets at the craps table?

- LLN says that the average winnings per bet would be close to -0.0141

Does it assure you that your losses will be small?

- No, you bet more = you lose more.

Does it assure you that if n is very large you will lose?

- Yes, LLN says that the average winnings will be close to -0.0141, which is negative.

T6

$Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx >= \int_{|X-\mu|\geq\epsilon} (x-\mu)^2 f(x)\,dx$

Since $|X-\mu| \geq \epsilon$ -> $(x-mu)^2 >= \epsilon^2$

We get $\int_{|X-\mu|\geq\epsilon} (x-\mu)^2 f(x)\,dx \geq \int_{|X-\mu|>=\epsilon} \epsilon^2 f(x)\,dx = \epsilon^2 P(|X-\mu| \geq \epsilon)$

So, $P(|X-\mu| >= \epsilon) \leq \frac{Var(X)}{\epsilon^2}$

T7

Chebyshev Inequality : $P(|X-\mu| >= \epsilon) \leq Var(X)/\epsilon^2$

$P(|X-\mu| >= 1.5SD) \leq \frac{SD^2}{(1.5SD)^2}$

$P(|X-\mu| >= 1.5SD) \leq 0.444$

The maximum possible fraction of data that deviates by more than 1.5 SD from the mean is 0.444

T8

```
# T8
1-binom.cdf(36000, 100000, 0.35)
```

```
1.804401073002282e-11
```

T9

CLT says that for large n, $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$

and $S_n \sim N(n\mu, n\sigma^2)$

For this problem, each X_i is from a Bernoulli distribution: $\mu = 0.5 * 0.7 = 0.35$ and $\sigma^2 = 0.35 * (1-0.35)$

```
# T9
1-norm.cdf(0.36,0.35,(0.35*0.65/100000)**0.5)
```

```
1.6791679158245643e-11
```

```
# or
1-norm.cdf(36000,35000,(0.35*0.65*100000)**0.5)
```

```
1.6791679158245643e-11
```

T10

$P(sum\ of\ score = 6) = P(A=0)P(B=6) + P(A=1)P(B=5) + \ldots + P(A=6)P(B=0)$

```
prob = 0
for i in range(7):
    a = poisson.pmf(i,3)
    b = poisson.pmf(6-i,2)
    prob += a*b
prob
```

```
0.14622280813987562
```

T11

$$
\begin{aligned}
P(Z=k) &= P(X+Y=k) \\
&= \sum_{i=0}^{k} P(X=i)P(Y=k-i) \\
&= \sum_{i=0}^{k} \frac{a^i e^{-a}}{i!} \cdot \frac{b^{k-i} e^{-b}}{(k-i)!} \\
&= \frac{e^{-(a+b)}}{k!} \sum_{i=0}^{k} \frac{k!}{i!(k-i)!} a^i b^{k-i} \\
&= \frac{e^{-(a+b)}}{k!} (a+b)^k \\
&= Poisson(a+b)
\end{aligned}
$$

if you don't know where $(a+b)^k$ comes from, go check out https://en.wikipedia.org/wiki/Binomial_theorem

T12

From T11, distribution of total goals = Poisson(3*10+2*10+1*44) = Poisson(94)

So, the expected value of the total amount of goals is 94.

T13

```
# T13
count=0
for i in range(1000000):
    a = sum(poisson.rvs(3,size=10))
    b = sum(poisson.rvs(2,size=10))
    c = sum(poisson.rvs(1,size=44))
    if a+b+c ==64 : count+=1
print(count/1000000)
```

```
0.00022
```

```
# or from T11
sum(poisson.rvs(94,size=1000000)==64)/1000000
```

```
0.000223
```

T14

For large λ , $Poisson(\lambda)$ can be approximated as $N(\lambda, \lambda)$

in this case $Poisson(94) \approx N(94, 94)$

This could be explained with CLT, by looking at $Poisson(94)$ as the sum of 94 results from a $Poisson(1)$ distribution (using what we proved in T11)

```
# T14
norm.cdf(64.5,94,94**0.5) - norm.cdf(63.5,94,94**0.5)
```

```
0.0003442906837397299
```

T15

```
# find z for 90%
# 1000 is big enough for clt -> assume normal dist
z = norm.ppf(0.95,0,1) # 0.95 is from 0.5 + 0.45
z
```

```
1.6448536269514722
```

SD of 1 trial = $\sqrt{p*(1-p)} = \sqrt{0.35*0.65}$

clt says that SD of $\bar{x}$ (mean from 1000 trial) = $\frac{\sqrt{0.35*0.65}}{\sqrt{1000}} = 0.01508$

with 90% confidence, we can say that $\mu \pm 1.645SD$ or $\mu \pm 0.0253$

So we could estimate the 90% confidence interval of the popularity of choice A as 0.35 ± 0.0253 or $35\% \pm 2.53\%$